

Notes: Comin, Lashkari & Mestieri (Econometrica, 2021)

Structural Change With Long-Run Income and Price Effects

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1 Big picture

- **Question.** What drives long-run structural transformation across agriculture, manufacturing, and services: changes in relative prices, nonhomothetic income effects, or both?
- **Core mechanism.** The paper introduces nonhomothetic CES preferences. These preserve constant substitution elasticities across sectors while allowing sectoral income elasticities to differ permanently across income levels.
- **Main empirical result.** Agriculture behaves as a necessity relative to manufacturing, services behave as a luxury, and manufacturing lies between the two. These patterns are stable across U.S. household data, Indian household data, and postwar cross-country data.
- **Decomposition result.** Income effects explain most within-country structural change: roughly 98% of the within-country variation for agriculture relative to manufacturing and 84% for services relative to manufacturing.
- **Economic takeaway.** Demand-side nonhomotheticity is not a short-run or low-income phenomenon that fades away. It remains important over long horizons and across countries.

2 Incremental contribution of the paper

- **Relative to classic structural-change models.** The paper integrates demand-side nonhomotheticity with supply-side productivity and relative-price mechanisms inside one tractable growth model.
- **Relative to Stone–Geary and PIGL.** Existing nonhomothetic demand systems often imply quickly flattening Engel curves or strong links between price and income effects. Nonhomothetic CES keeps price elasticities constant while allowing stable long-run income effects.
- **Relative to empirical measurement.** The paper derives a real consumption index implied by the model and uses it to estimate the demand system in micro and macro data.
- **Relative to decomposition exercises.** The log-linear structure separates relative-price effects from real-consumption effects, allowing a clean income-versus-price accounting of structural transformation.
- **Relative to growth theory.** The model shows how structural change affects capital accumulation and real interest-rate dynamics, not just sectoral shares.

Interpretation: The incremental contribution is not only a new aggregator. The paper shows that the aggregator is estimable, fits household and cross-country data, and changes the quantitative interpretation of structural transformation.

3 Data and measurement

3.1 Sectoral aggregation

The empirical work focuses on agriculture, manufacturing, and services, with manufacturing usually used as the base sector. The paper also extends the framework to more sectors by separating housing from services and by studying service splits.

Interpretation: The three-sector structure matches the main structural-change literature while keeping the demand system tractable.

3.2 U.S. CEX data

The U.S. micro estimates use the Consumer Expenditure Survey. Household expenditures are grouped into agriculture, manufacturing, and services. The regressions control for household characteristics and, in some specifications, region and year-by-quarter fixed effects.

Interpretation: CEX data let the authors estimate Engel slopes and price effects at the household level, rather than relying only on aggregate country variation.

3.3 Indian NSS data

The Indian micro estimates use National Sample Survey expenditure data. India provides a very different income distribution and sectoral composition from the United States.

Interpretation: The India exercise is an external validity test. Similar relative nonhomotheticity estimates across India and the United States support the stability of the demand system.

3.4 Cross-country panel

The macro estimates use historical aggregate-level panel data for 39 countries during the postwar period. Outcomes include sectoral employment shares and value-added shares.

Interpretation: These data allow the paper to study structural transformation directly: the movement of employment and value added across sectors as countries grow.

3.5 Key measured objects

The key objects are relative expenditure or employment shares, relative prices, the nonhomothetic CES real consumption index, and sector-specific nonhomotheticity parameters.

$$\log C_t = \log \left(\frac{E_t}{p_{bt}} \right) + \frac{1}{1 - \sigma} \log \omega_{bt}.$$

Interpretation: The real consumption index is the bridge between theory and data. It allows relative demand to be written as a function of prices and real consumption.

4 Models and empirical specifications

4.1 Relative demand

$$\log \left(\frac{\omega_i}{\omega_j} \right) = (1 - \sigma) \log \left(\frac{p_i}{p_j} \right) + (1 - \sigma)(\epsilon_i - \epsilon_j) \log C + \log \left(\frac{\Omega_i}{\Omega_j} \right).$$

Interpretation: The first term is the price effect and the second term is the income effect. The constant elasticity of substitution is preserved while income elasticities vary by sector.

4.2 Income elasticity

$$\eta_i = \frac{\partial \log C_i}{\partial \log E} = \sigma + (1 - \sigma) \frac{\epsilon_i}{\bar{\epsilon}}, \quad \bar{\epsilon} = \sum_i \omega_i \epsilon_i.$$

Interpretation: For $\sigma < 1$, sectors with higher ϵ_i are more income elastic. Agriculture has low ϵ_i and services have high ϵ_i .

4.3 Aggregate estimating equation

$$\log \left(\frac{L_{it}}{L_{mt}} \right) = (1 - \sigma) \log \left(\frac{p_{it}}{p_{mt}} \right) + (1 - \sigma)(\epsilon_i - 1) \log C_t + (\epsilon_i - 1) \log \omega_{mt} + \xi_i + \nu_{it}.$$

Interpretation: This equation separates sectoral employment-share movements into relative-price and income-effect components.

4.4 Variance decomposition

$$\begin{aligned} \text{Price contribution} &= \frac{\text{Var}[(1 - \hat{\sigma}) \log(p_{it}/p_{mt})]}{\text{Var}[\log(L_{it}/L_{mt}) - \hat{\xi}_i]}, \\ \text{Income contribution} &= \frac{\text{Var}[(1 - \hat{\sigma})(\hat{\epsilon}_i - 1) \log C_t]}{\text{Var}[\log(L_{it}/L_{mt}) - \hat{\xi}_i]}. \end{aligned}$$

Interpretation: The decomposition is the paper's main quantitative accounting exercise.

5 Identification and empirical/theoretical design

- **Demand-system identification.** Relative price variation identifies σ , while the slope of relative sectoral demand with respect to real consumption identifies relative nonhomotheticity.
- **Normalization.** Only relative nonhomotheticities are identified, so the paper normalizes manufacturing to one and checks alternative bases.
- **Micro-to-macro validation.** The same demand system is estimated in U.S. CEX data, Indian NSS data, and aggregate cross-country data.
- **Controls.** The macro specifications include country-sector fixed effects and trade controls; the micro specifications include household controls and fixed effects.
- **Model comparison.** The paper compares the fit of nonhomothetic CES with Stone–Geary and PIGL preferences.
- **Robustness.** The paper studies alternative bases, value-added instead of final expenditure, India versus the United States, OECD versus non-OECD samples, and more than three sectors.

Detailed interpretation: The design is a structural identification strategy rather than a single natural experiment. Its credibility comes from the model's sharp log-linear restrictions and from the consistency of estimates across very different datasets.

How it fits the paper: The crucial empirical claim is that income effects remain stable over long horizons. Tables I, II, III, V, and VI all support this stability.

6 Tables and figures

6.1 Figure 1: Partial correlations of sectoral expenditure and aggregate consumption; Table I: Estimates, CEX Final-Good Expenditure

Read: Figure 1 shows agriculture expenditure falling and services expenditure rising with aggregate income after controls. Table I estimates agriculture as a necessity, services as a luxury, and $\sigma < 1$.

Detailed interpretation: Figure 1 motivates the paper, while Table I provides the structural micro estimate. Together they show that relative Engel slopes are stable and economically large.

Economic message: The demand-side channel is visible before imposing the full model.

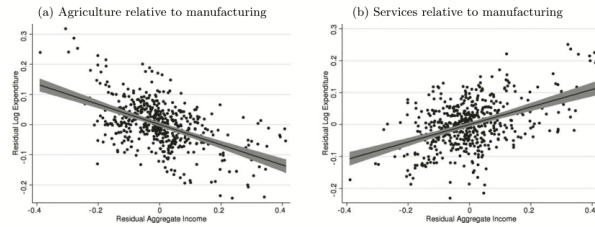


FIGURE 1.—Partial correlations of sectoral expenditure and aggregate consumption. Notes: Data for OECD countries, 1970–2005. Each point corresponds to a country-year observation after partialling-out sectoral prices and country fixed effects. The red line depicts the OLS fit, the shaded regions, the 95% confidence interval.

consumption through price (substitution) effects (Baumol (1967), Ngai and Pissarides

(a) Figure 1: residual sectoral expenditure against residual income.

	(1)	(2)	(3)	(4)	(5)	(6)
σ	0.26 (0.04)	0.28 (0.04)	0.28 (0.03)	0.20 (0.05)	0.31 (0.04)	0.33 (0.05)
$\epsilon_a - 1$	-0.80 (0.06)	-0.83 (0.07)	-0.81 (0.06)	-0.70 (0.07)	-0.95 (0.09)	-0.97 (0.10)
$\epsilon_s - 1$	0.65 (0.07)	0.68 (0.07)	0.75 (0.06)	0.67 (0.07)	0.82 (0.09)	0.85 (0.10)
Expenditure Re-Weighted	N	Y	N	Y	N	Y
Region FE	N	N	Y	Y	Y	Y
Year $\times$ Quarter FE	N	N	N	N	Y	Y

^aAll regressions include household controls (described in the text). Standard errors clustered at the household level shown in parentheses. The number of observations is 60,925 in all regressions.

³³We also note that our estimation strategy is different from the one proposed in Hanoch (1975). Hanoch proposed an estimation based on double differences that can only identify $l - 2$ nonhomotheticity parameters.

³⁴We thank the editor for this suggestion.

(b) Table I: CEX estimates.

6.2 Table II: Sample Splits, CEX Final-Good Expenditure; Figure 2: Partial correlation of the real consumption index and relative log expenditure shares

Read: Table II shows that the estimated slopes are stable across income groups and periods. Figure 2 shows that relative expenditure shares are approximately log-linear in the constructed real consumption index.

Detailed interpretation: These results address the concern that nonhomotheticity vanishes as households become richer.

Economic message: Income effects are long-run forces, not just low-income transitional effects.

TABLE II
SAMPLE SPLITS, CEX FINAL-GOOD EXPENDITURE^a

	<P50 (1)	>P50 (2)	Pre'05 (3)	Post'05 (4)
σ	0.35 (0.07)	0.31 (0.05)	0.33 (0.07)	0.25 (0.05)
$\epsilon_a - 1$	-0.89 (0.17)	-0.99 (0.12)	-0.98 (0.15)	-0.92 (0.08)
$\epsilon_s - 1$	0.75 (0.19)	0.59 (0.16)	0.74 (0.14)	0.65 (0.10)

^aRegressions estimated using CEX-replicate weights. Households controls included in all regressions (as described in the main text). All regressions include Region and Year $\times$ Quarter fixed effects. Standard errors clustered at the household level. The estimations in columns (2) and (3) are performed imposing the constraint $\epsilon_a \geq 0$ (by estimating an exponential transformation of the variable). The corresponding standard errors are computed using the delta method.

(a) Table II: sample splits.

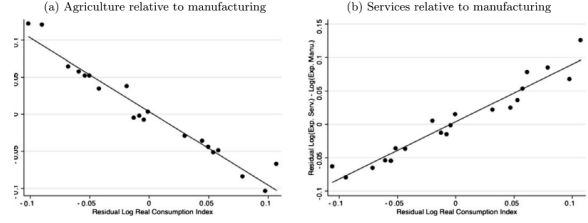


FIGURE 2.—Partial correlation of the real consumption index and relative log expenditure shares, CEX. Notes: These plots depict the (binned) residuals corresponding to the average value of 20 equal-sized bins of data. The red line depicts the linear regression between the residualized variables.

(b) Figure 2: real consumption index and relative shares.

6.3 Table III: Cross-Country Estimates, $\epsilon_m = 1$; Table IV: Accounting for Structural Change, Baseline Estimates

Read: Table III estimates similar nonhomotheticity rankings in world, OECD, and non-OECD samples. Table IV reports that income effects explain 98% of agriculture/manufacturing within-country variation and 84% of services/manufacturing variation.

Detailed interpretation: This is the central quantitative result: structural transformation is mostly an income-effect phenomenon in the estimated model.

TABLE III
CROSS-COUNTRY ESTIMATES, $\epsilon_m = 1$ ^a

	World		OECD		Non-OECD	
	(1)	(2)	(3)	(4)	(5)	(6)
σ	0.57 [0.32, 0.69]	0.50 [0.26, 0.71]	0.25 [0.20, 0.66]	0.35 [0.03, 0.55]	0.63 [0.06, 0.74]	0.48 [0.34, 0.75]
$\epsilon_a - 1$	-0.98 [-1.13, -0.41]	-0.89 [-1.14, -0.46]	-0.99 [-1.00, -0.38]	-0.99 [-1.00, -0.66]	-0.91 [-1.15, -0.58]	-0.80 [-1.14, -0.40]
$\epsilon_s - 1$	0.17 [0.07, 0.60]	0.21 [0.03, 0.67]	0.27 [0.03, 0.55]	0.25 [0.09, 1.95]	0.18 [0.11, 2.08]	0.37 [0.03, 0.67]
Country $\times$ Sector FE	Y	Y	Y	Y	Y	Y
Trade Controls	N	Y	N	Y	N	Y
Observations	1626	1626	492	492	1134	1134

^aBootstrapped 95% confidence intervals clustering at the country level shown in square brackets (computed through bootstrapping 50 samples with replacement). The estimations in columns (3) and (4) are performed by imposing the constraint that $\epsilon_a \geq 0$ (by estimating an exponential transformation of the variable).⁴³

(a) Table III: cross-country estimates.

TABLE IV
ACCOUNTING FOR STRUCTURAL CHANGE, BASELINE ESTIMATES^a

	log($\frac{\text{Agriculture}}{\text{Manufacturing}}$)	log($\frac{\text{Services}}{\text{Manufacturing}}$)
Explained over Total Variance	0.97	0.57
Within over Explained Variance	0.46	0.61
Within-Country Variance Decomposition		
Price Effects	0.02	0.27
Income Effects	0.98	0.84
Both Effects	1.00	1.00

^aExplained over Total Variance is computed as $\text{Var}[\log(\frac{L^n_{it}}{T^n_{it}})] / \text{Var}[\log(\frac{L^n_{it}}{T^n_{it}})]$. Within over Explained Variance is computed as $\text{Var}[(1 - \hat{\sigma}) \log(\frac{L^n_{it}}{P^n_{it}}) + (1 - \hat{\sigma})(\hat{\sigma} - 1) \log(\frac{L^n_{it}}{C^n_{it}})] / \text{Var}[\log(\frac{L^n_{it}}{T^n_{it}})]$.

(b) Table IV: variance decomposition.

6.4 Table V: Variance Decomposition Using Alternative Bases; Table VI: Baseline Regression for India, NSS Expenditure

Read: Table V shows robustness to base-sector normalization. Table VI shows that Indian NSS estimates generate ratios close to U.S. estimates.

Detailed interpretation: These are robustness and external-validity tests. They support the claim that the preference parameters are stable across normalizations and very different income distributions.

TABLE V
VARIANCE DECOMPOSITION USING ALTERNATIVE BASES^a

	Estimates Under Alternative Bases (see Table IX)					
	Agri. as base	Manu. as base	Serv. as base	3 sec. as base		
σ	0.33 (0.03)	0.57 (0.07)	0.33 (0.03)	0.40 (0.04)		
$\epsilon_a/\epsilon_m - 1$	-0.67 (0.06)	-0.98 (0.06)	-0.86 (0.07)	-0.68 (0.04)		
$\epsilon_s/\epsilon_m - 1$	0.26 (0.05)	0.17 (0.04)	0.23 (0.05)	0.31 (0.06)		
Variance Decomposition Using Alternative Bases						
	$\log(\frac{\text{Agriculture}}{\text{Manufacturing}})$			$\log(\frac{\text{Services}}{\text{Manufacturing}})$		
	A-base	M-base	S-base	A-base	M-base	S-base
Price Effects	0.03	0.02	0.02	0.10	0.27	0.16
Income Effects	0.94	0.98	0.94	0.95	0.84	1.00

^a Column A-base reports the result of performing the variance decomposition in Equation (42) using the estimates obtained with agriculture as the base sector (reported in the first three rows of column "Agri. as base" in this table). Analogously, M-base, S-base, and 3-base denote the variance decomposition results using the estimates obtained with manufacturing, services, and the linear combination of the 3 sectors as bases. Robust standard errors shown in parentheses.

(a) Table V: alternative bases.

TABLE VI
BASELINE REGRESSION FOR INDIA, NSS EXPENDITURE^a

	(1)	(2)	<P50 (3)	>P50 (4)	Only Urban (5)	U.S. (6)	U.S. (7)
$\bar{\epsilon}_a - \bar{\epsilon}_m$	-0.63 (0.05)	-0.55 (0.07)	-0.62 (0.05)	-0.69 (0.29)	-0.57 (0.05)	-0.52 (0.07)	-0.61 (0.09)
$\bar{\epsilon}_s - \bar{\epsilon}_m$	0.49 (0.07)	0.42 (0.10)	0.69 (0.53)	0.51 (0.08)	0.56 (0.08)	0.45 (0.11)	0.49 (0.09)
$(\bar{\epsilon}_s - \bar{\epsilon}_m)/(\bar{\epsilon}_a - \bar{\epsilon}_m)$	-0.78	-0.76	-1.11	-0.74	-0.98	-0.87	-0.80
$(\epsilon_s - 1)/(\epsilon_a - 1)$, baseline estimates	-0.81	-0.82	-0.84	-0.60	-0.81	-0.81	-0.81
Expenditure Re-Weighted	N	Y	N	N	N	Y	N
Time $\times$ Region $\times$ Inc. Quintile FE	Y	Y	Y	Y	Y	Y	Y
T. $\times$ Reg. $\times$ Inc. Quint. $\times$ Rural FE	Y	Y	Y	Y	N	N	N

(b) Table VI: India NSS estimates.

6.5 Figure 3: Comparison of demand systems and fit for Taiwan; Table VII: Correlation of Nominal and Real Value Added

Read: Figure 3 compares nonhomothetic CES, Stone–Geary, and PIGL. Nonhomothetic CES has the best within fit. Table VII shows that model-predicted nominal/real correlations are close to the data.

Detailed interpretation: The Taiwan exercise is a visual horse race showing why the model improves on previous preferences.

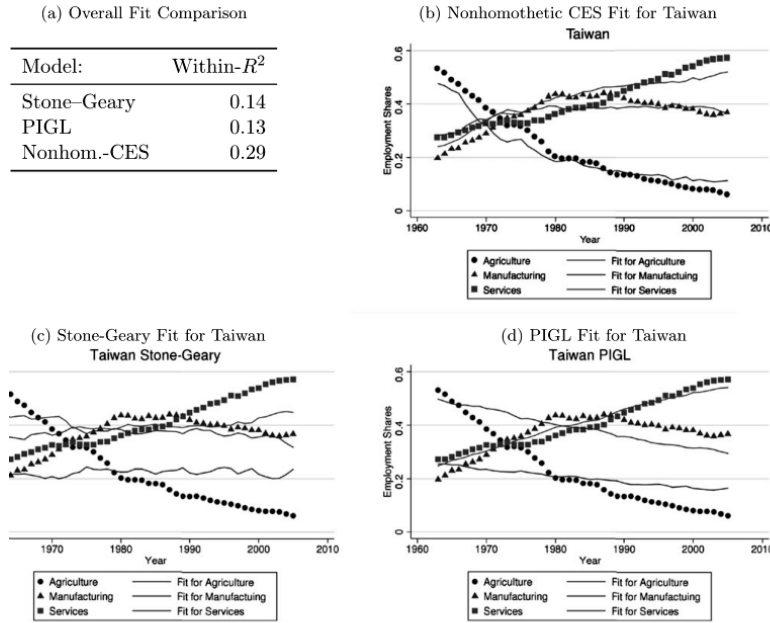


FIGURE 3.—Comparison of demand systems and fit for Taiwan.

motheticity parameters appear with the expected signs and overall similar magnitudes in the baseline regression. The price elasticity appears to be somewhat smaller.

(a) Figure 3: demand-system fit comparison for Taiwan.

TABLE VII
CORRELATION OF NOMINAL AND REAL VALUE ADDED^a

	Data	Model
Agriculture/Manufacturing	0.96	0.94
Services/Manufacturing	0.87	0.71

^a Model-predicted values constructed using the structural estimates

(b) Table VII: nominal and real value-added correlation.

6.6 Figure 4: Partial correlations of sectoral value added and the real consumption index; Figure 5: Real income per capita from PWT and nonhomothetic CES index

Read: Figure 4 shows the same Engel pattern for value-added shares. Figure 5 shows that the model's real consumption index is closely related to PWT real income per capita.

Detailed interpretation: These figures reduce concern that the results depend on final-good expenditure measurement.

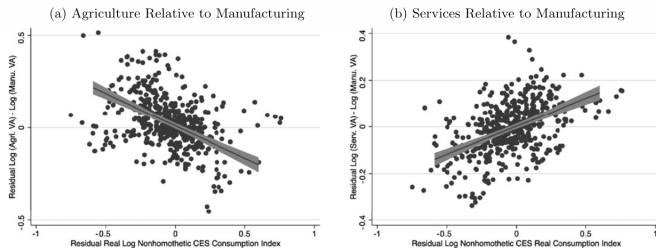


FIGURE 4.—Partial correlations of sectoral value added and the real consumption index. Notes: Data for OECD Countries, 1970–2005. Each variable has been residualized by partialling out country fixed effects and relative prices. Each point represents a country-year observation. The red line depicts the OLS fit, the shaded regions, the 95% confidence interval.

(a) Figure 4: value added and real consumption index.

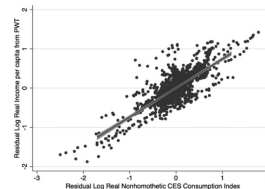


FIGURE 5.—Real income per capita from PWT and nonhomothetic CES index. Note: Both measures are residualized by partialling out country fixed effects.

(b) Figure 5: PWT income and nonhomothetic CES index.

6.7 Table VIII: Model Parameters for the Calibration Exercise; Figure 6: Transitional dynamics: comparison with Neoclassical Growth Model

Read: Table VIII lists calibrated preference and technology parameters. Figure 6 compares the full model with benchmark growth models.

Detailed interpretation: Structural change slows capital accumulation and interest-rate convergence relative to the neoclassical benchmark, with nonhomotheticity playing the larger role.

TABLE VIII
MODEL PARAMETERS FOR THE CALIBRATION EXERCISE

Parameter	γ_a	$\gamma_m = \gamma_0$	γ_s	σ	ϵ_a	ϵ_m	ϵ_s	α	θ	β	δ
Value	0.029	0.013	0.011	0.50	0.05	1.00	1.20	0.33	2.20	0.96	0.10

Table VIII: calibration parameters.

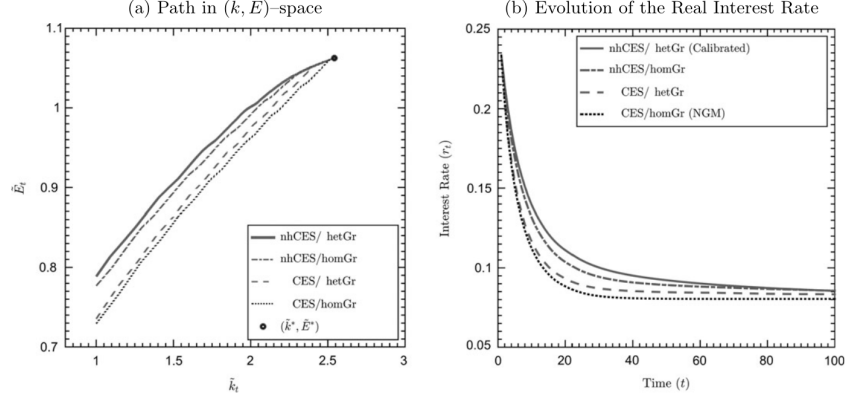


FIGURE 6.—Transitional dynamics: comparison with Neoclassical Growth Model (NGM). The evolution of the economy starting from initial per capita stock of capital of $\bar{k}_0 = 1 < \bar{k}^* = 2.50$. The parameters for the Calibrated Model are given in Table VIII. The nhCES/hetGr corresponds to the calibrated model with nonhomothetic CES and heterogeneous rates of sectoral productivity growth. The nhCES/homGr model corresponds to the case with nonhomothetic CES preferences and homogeneous rates of sectoral productivity growth, $\gamma_i = 0.011$ for $i \in \{a, m, s\}$. The CES/hetGr model corresponds to the case with homothetic CES preferences, $\epsilon_i = 1.20$ for $i \in \{a, m, s\}$. The CES/homGr corresponds to the case of the Neoclassical Growth Model (NGM) where both the rates of productivity growth and the nonhomotheticity parameters are homogeneous across sectors. $\bar{k}^*$ and $\bar{E}^*$ denote the asymptotic normalized per capita stock of capital and total consumption expenditure, respectively.

Figure 6: transitional dynamics and interest rates.

7 Short summary

- The paper introduces nonhomothetic CES preferences into growth theory.
- The model separates income effects from price effects in relative sectoral demand.
- U.S. CEX, Indian NSS, and cross-country data all support stable sectoral nonhomotheticity.
- Agriculture is a necessity, services are a luxury, and manufacturing is intermediate.
- Income effects account for the bulk of within-country structural transformation.
- The model fits better than Stone–Geary and PIGL alternatives in the Taiwan comparison.
- Structural change affects aggregate transition dynamics, including capital accumulation and interest rates.

8 Conclusion

8.1 Core conclusion

Comin, Lashkari, and Mestieri show that a tractable nonhomothetic CES demand system can explain long-run structural transformation. Income effects remain strong across income levels and account for most within-country reallocation.

8.2 Economic intuition

As real consumption rises, households move expenditure away from low-income-elastic sectors and toward high-income-elastic sectors. Relative prices still matter, but the paper finds that income effects dominate the long-run movement in sectoral shares.

8.3 Why the incremental contribution matters

The paper makes nonhomothetic demand as tractable as CES while retaining empirically meaningful Engel curves. This allows applied general equilibrium models to use a flexible demand system without abandoning CES-style substitution structure.

8.4 Policy and macro implications

Structural transformation is not only a consequence of productivity and relative prices. Growth itself changes demand composition, which can affect sectoral employment, capital accumulation, and real interest-rate dynamics.

8.5 Limitations and open questions

The paper's decomposition depends on the maintained nonhomothetic CES structure and on sectoral price measurement. More disaggregated applications may require richer sector definitions and additional heterogeneity.

8.6 Takeaway

Nonhomotheticity is a long-run force. Once it is modeled in a tractable way, demand-side income effects become central to the economics of structural transformation.